

ExpressLRS

Firmware Rev. 3.3.0 (ae9df3)

Load target configuration (remember to press "Save Target Configuration" below):

Choose File no file selected

or drop files here

CRSF Serial Pins

RX pin



13

Pin used to
receive CRSF
signal from the
handset

TX pin



13

Pin used to
transmit CRSF
telemetry to the

Radio Chip Pins & Options

BUSY pin



DIO0 pin



DIO1 pin



DIO2 pin



MISO pin



MOSI pin



NSS pin



RST pin



SCK pin



BUSY_2 pin

handset (may be the same as the RX PIN)

GPIO Input connected to SX128x busy pin

Unused on SX128x, Interrupt pin for SX127x

Interrupt pin for SX128x

Unused on SX128x and 127x

MISO connected to (possibly) multiple SX1280/127x

MOSI connected to (possibly) multiple SX1280/127x

Chip select pin for first SX1280/127x

Reset pin connected to (possibly) multiple SX1280/127x

Clock pin connected to (possibly) multiple SX1280/127x

Busy pin for



DIO0_2 pin



DIO1_2 pin



NSS_2 pin

DCDC
enabled☐RFO_HF
enabled☐

Radio Antenna

CTRL pin

CTRL_COMPL
pin

Radio Power

PA enable pin



APC1 pin



APC2 pin



second SX1280

Interrupt pin for
second SX127xInterrupt pin for
second SX1280Chip select pin
for second
SX1280Use the SX1280
DC-DC converter
rather than LDO
voltage regulator
(15uH inductor
must be present)SX127x PA to
use, either the
RFO_HF or
PA_BOOST
(depends on
circuit design)Pin connected to
Antenna select
pin on power
amplifierInverted CTRL
for devices using
antenna
selectors that
need separate
pins for A/B
selectionEnable the
power amplifier
(active high)Power amplifier
control voltage

Power amplifier

RXEN pin	⬆️⬆️ 27	control voltage
TXEN pin	⬆️⬆️ 26	Enable RX mode LNA (active high)
RXEN_2 pin	⬆️⬆️	Enable TX mode PA (active high)
TXEN_2 pin	⬆️⬆️	Enable RX mode LNA on second SX1280 (active high)
Min Power	25mW ⬆️⬆️	Enable TX mode PA on second SX1280 (active high)
High Power	1000mW ⬆️⬆️	Minimum selectable power output
Max Power	1000mW ⬆️⬆️	Highest selectable power output (if option for higher power is NOT enabled)
Default Power	50mW ⬆️⬆️	Absolute maximum selectable power output (only available if 'higher power' option is enabled)
Power Level control	via SEMTECH ⬆️⬆️	Default power output when resetting or first flashing a module
Level Value(s)	-18,-15,-12,-7,-4,2	How the power level is set
		Comma- separated list of values that set

Radio Power Detection

PDET pin 

Intercept

Slope

Analog Joystick

ADC pin 

Values

Digital Joystick

Pin 1 

the power output

Analog input (up to 1.1V) connected to 'power detect' pin on PA for adjustment of the power output Intercept and Slope are used together to calculate the dBm from the measured mV on the PDET pin $\text{dBm} = \text{mV} * \text{slope} + \text{intercept}$, this is then used to adjust the actual output power accordingly

Analog Input (3.3V max) use to read joystick direction using a resistor network Comma-separated list of ADC values (12-bit) for UP, DOWN, LEFT, RIGHT, ENTER, IDLE

These 3 pins create a binary value for the

Pin 2  Pin 3  **Mood Lighting**RGB LED pin  RGB LED is GRB ☒RGB indexes for Status RGB indexes for Boot animation LED pin  Red LED pin  Red LED inverted ☐Green LED pin  Green LED inverted ☐

Blue LED pin

joystick direction

7 = IDLE, 6 =
OK, 5 = DOWN
4 = RIGHT, 3 =
UP, 2 = LEFT

Signal pin for
WS2812 RGB
LED or LED strip

Most WS2812
RGB LEDs are
actually GRB

Indexes into the
"string" of RGB
LEDs (if empty
then only LED at
0 is used)

Indexes into the
"string" of RGB
LEDs (if empty
status indexes
are used)

Only use when
only a single
LED is used

If there are
multiple LEDs,
then this is the
pin for the RED
LED

LEDs are active
LOW unless this
is checked

If there is a
GREEN LED as
well as RED
above

Check if the LED
is active HIGH

Pin for a 3rd,

⬆️	BLUE, LED!
Blue LED inverted ☐	Check if the LED is active HIGH
Button(s)	
Button 1 pin ⬆️	Single/first (active low) button
Button 1 RGB Index	Index of button LED in RGB string, leave empty for no RGB LED
Button 2 pin ⬆️	Second (active low) button
Button 2 RGB Index	Index of button LED in RGB string, leave empty for no RGB LED
OLED/TFT (Crotch TV)	
Screen type I2C OLED (SSD1306 128x64) ⌵	Type of OLED connected
180 rotation ☒	Select to rotate the display 180 degrees
CS pin ⬆️	Chip Select (if using SPI)
DC pin ⬆️	Data/Command Select (if using SPI)
MOSI pin ⬆️	Data (if using SPI)
RST pin ⬆️	Reset
SCK pin ⬆️ 32	Clock (either SPI or I2C)
SDA pin ⬆️ ⬇️ 22	Data (I2C)

BL pin

**Backpack /
Logging**Enable
Backpack

Baud Rate

460800

RX pin



3

TX pin



1

BOOT pin



EN pin

Passthrough
baud**I2C & Misc Devices**

SCL pin

**Backlight**If a TX backpack
is connectedBaud rate used
to communicate
to the backpack
(normally
460800)Connected to TX
pin on backpackConnected to RX
pin on backpackPin connected to
GPIO0 pin on
backpack
ESP8285, allows
passthrough
flashingPin connected to
EN pin on
backpack
ESP8285, allows
passthrough
flashingBaud rate to
flash the
backpack
ESP8285
(default is to use
the baud rate
above)I2C clock pin
used to
communicate
with I2C devices
(may be the
same as OLED)

SDA pin  

Fan enable pin 

Fan PWM pin 

Fan PWM output values

Fan TACHO pin 

Has STK8xxx G-sensor ☐

G-sensor interrupt pin 

Has LM75A Thermal sensor ☐

I2C)

I2C data pin used to communicate with I2C devices (may be the same as OLED I2C)

Pin used to enable a cooling FAN (active HIGH)

If the fan is controlled by PWM

If the fan is PWM controlled, then this is the list of values for the PWM output for the matching power output levels

If the fan has a "tachometer" interrupt pin

Checked if there is a STK8xxx g-sensor on the I2C bus

Pin connected the STK8xxx g-sensor for interrupts

Checked if there is a LM75A thermal sensor on the I2C bus

SAVE TARGET CONFIGURATION